Does deep cross-modal fusion improve prediction of late-adolescent psychopathology? A comparison with mixed-effects models in a Swedish twin cohort

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Abstract

Background: Deep multimodal fusion networks can model interactions between psychiatric risk domains, but whether they predict better than well-specified regression models is rarely tested on identical data. We compared a deep cross-modal fusion network, in which adolescent substance use is fused with each other risk domain, against mixed-effects models with matched inputs and interactions. **Methods:** We analysed 4,033 twins (2,690 pairs) from the Child and Adolescent Twin Study in Sweden with complete data on 105 predictors in 11 groups, including substance use at age 15, neurodevelopmental traits, adverse experiences and 16 cell-type-specific polygenic scores. Outcomes were continuous scores of psychotic-like and hypomanic experiences and of depressive symptoms at age 18. Comparators were a linear and a generalised additive mixed model, each with a twin-pair random intercept and 24 substance-use interaction terms mirroring the network's fusion design. All models were evaluated on the same five twin-pair-grouped splits, with corrected resampled t-tests. We also assessed calibration, performance by sex, models without age-18 predictors, and whether the network's attention gates agreed with SHAP values and permutation importance. **Results:** For psychotic-like and hypomanic experiences, mean test R^2 was 0.206 (SD 0.019) for the network, 0.228 (0.016) for the linear mixed model and 0.227 (0.015) for the additive mixed model. For depressive symptoms, it was 0.198 (0.021), 0.202 (0.022) and 0.200 (0.023). No paired difference was statistically significant. A cross-validated lasso with default settings reached 0.261 and 0.234. Of the 24 interaction terms, three and one had unadjusted $p < 0.05$, about the number expected by chance, and none remained significant after correction. [TODO: one sentence on calibration, sex and age-15-only results — prediction_evaluation.py] [TODO: one sentence on gate–SHAP–permutation agreement — attribution_agreement.py]

Conclusions: In this twin cohort, the fusion network did not predict better than mixed-effects models with matched inputs and interactions, even though only the network was tuned and part of its configuration was chosen using test data. The data gave little evidence of interactions between adolescent substance use and other risk domains. Deep psychiatric prediction models should be benchmarked against well-specified regression models on identical splits before their complexity or attention-based explanations are relied on.

Keywords: Psychotic-like experiences, Depressive symptoms, Adolescent substance use, Polygenic risk scores, Deep learning, Multimodal fusion, Mixed-effects models, Prediction model, Twin study

1 Background

Psychotic-like experiences, hypomanic symptoms and depressive symptoms are common in late adolescence. Most are transient, but they are associated with later psychiatric morbidity and help-seeking [CITATION NEEDED: adolescent psychotic-like experiences/subclinical symptoms and later psychopathology; candidate Kelleher/Cederlöf et al. 2017, Schizophr Res, doi 10.1016/j.schres.2016.08.029 — content unverified]. Their known correlates span several domains: adolescent substance use, particularly cannabis [1, 2] [VERIFY: content of R-DIFORTI beyond psychotic-disorder incidence]; childhood adversity [3]; socioeconomic circumstances [4] [VERIFY: content]; and inherited liability captured by polygenic risk scores (PRS) [5] [VERIFY: content]. Mapping psychiatric genetic signal onto specific brain cell types [6] [VERIFY: content] now allows cell-type-specific PRS, which may carry more targeted information than a genome-wide score.

A long-standing hypothesis holds that these domains act jointly rather than additively, for example that genetic liability or early adversity changes the effect of substance use [7] [VERIFY: content]. If such interactions are strong, models that represent them should predict symptoms better than additive models. Deep multimodal fusion networks are one way to do this: they combine groups of inputs ("modalities") through learned non-linear interactions, and they are increasingly applied to heterogeneous health data [8]. A fusion architecture in which one exposure domain conditions the representation of every other domain, such as the iterative gated fusion of the Deep Cross-Modal Fusion Network (DCMFNet) [9], matches the interaction hypothesis directly. Such networks also contain attention-like gates, which are often read as indicators of which inputs matter.

Matching the hypothesis does not guarantee better predictions. In clinical prediction, a systematic review found no performance benefit of machine learning over logistic regression in comparisons at low risk of bias, and calibration was rarely assessed [10]. On medium-sized tabular data, tree ensembles usually outperform deep networks [11]. Earlier machine-learning work in the cohort used here found performance comparable to regression for adolescent mental-health problems [12] [VERIFY: read abstract/full text before citing this finding]. Comparisons between model classes are

also prone to artefacts that favour the more flexible model, including model selection on test data, feature screening outside the training data [13] [VERIFY: content], related individuals on both sides of a split [CITATION NEEDED: family/twin dependence and grouped cross-validation], unequal tuning effort, and comparators that do not contain the interactions the flexible model is credited with. Finally, attention weights are not necessarily faithful explanations of a model's predictions [14], so gate values need to be checked against attribution methods that measure what the model relies on.

We therefore compared DCMFNet, adapted so that substance use at age 15 is fused with each other risk domain, against linear and generalised additive mixed models given the same predictors, a twin-pair random intercept and interaction terms that mirror the network's fusion design. All models were evaluated on identical twin-pair-grouped splits in the Child and Adolescent Twin Study in Sweden [15], with outcomes at age 18. We asked (1) whether the fusion network predicted better than the mixed models, including calibration, performance by sex and models restricted to predictors measured by age 15; (2) whether the mixed models showed evidence of the substance-use interactions that motivate the fusion design; and (3) whether the network's attention gates agreed with SHAP values and permutation importance computed on the same models, and with exact SHAP values from the linear mixed model.

2 Methods

2.1 Study population

We used data from the Child and Adolescent Twin Study in Sweden (CATSS), a population-based cohort of twins born in Sweden and identified through the Swedish Twin Registry. Twins and their parents were invited to take part at ages 9 (or 12), 15 and 18 [15]. We used parent-reported neurodevelopmental measures from the age-9 assessment, self-reports at ages 15 and 18, and genotype data available for a subset of participants. [AUTHOR INPUT: birth cohorts included, data extraction date, who reported the age-9 measures]

Individuals were eligible if they had substance-use data at age 15 and outcome data at age 18. The analysis sample comprised all eligible individuals with complete data on every predictor and both outcomes: 4,033 individuals from 2,690 twin pairs. [TODO: participant flow (Fig. 1) from analysis_sample_flow.csv — reconcile with thesis Figs 3.1/3.3 (13,146 → 9,261 eligible; 9,160 genotyped), TODO W-02] We compared the analysis sample with excluded eligible participants on sex, outcomes, substance use at age 15 and summary scores of each predictor group, using standardised mean differences (Table 1). Counts below five are not reported.

2.2 Outcomes

Two continuous outcomes were derived from self-report items at age 18. [AUTHOR INPUT: instrument names, item response scales and references — TODO M-01]

- **Psychotic-like and hypomanic experiences (PLE-HM):** sum of 24 items, 7 on psychotic-like experiences (for example hearing voices or feeling spied upon) and 17 on hypomanic symptoms (for example racing thoughts or decreased need for sleep), divided by 46.
- **Depressive symptoms (DEP):** sum of 11 items (for example feeling sad, lonely or not enjoying life), divided by 33.

Both scores lie between 0 and 1 and were analysed as continuous outcomes without clinical thresholds.

2.3 Predictors

The 105 predictors were organised into 11 groups defined before analysis (number of variables in parentheses):

1. Substance use at age 15 (SUD15; 6): frequency of cigarette, snus, alcohol, cannabis and non-medical painkiller use, and the maximum across other drug items.
2. Polygenic risk scores (PRS; 16): a genome-wide schizophrenia PRS and cell-type-specific PRS restricted to genes specifically expressed in brain cell types of a human brain cell atlas [6, 16], computed with PRS-CS [17]. [AUTHOR INPUT: GWAS summary statistics, LD reference, PRS-CS settings, genotyping platforms, imputation and QC — TODO W-03]
3. Psychotic-like, hypomanic and emotional symptoms at age 15 (SCZ15; 23).
4. ADHD symptoms at age 9 (ADHD9; 19).
5. Autism-related traits at age 9 (ASD9; 17). [VERIFY: instrument (A-TAC?) and informant for 4–5]
6. Adverse experiences at age 15 (ACE15; 7), mainly bullying victimisation.
7. Adverse experiences at age 18 (ACE18; 4): emotional abuse, hate crime, witnessing crime and other abuse or violence.
8. Substance use at age 18 (SUD18; 4).
9. Parental socioeconomic indicators (SES; 4): parental education and country of birth.
10. Sex (1). [AUTHOR INPUT: register-based sex vs self-report (SAGER)]
11. Genotyping batch × genetic principal component terms (4): products of batch indicators and the first two principal components, to adjust for population structure and batch effects.

ACE18 and SUD18 were measured at the same assessment as the outcomes, so models that include them combine prospective and concurrent information. We therefore repeated all analyses without these groups (see *Sensitivity analyses*).

Variable screening. From 236 candidate variables, we removed variables with more than 90% missing values, combined or removed variables with pairwise $|r| > 0.9$, removed variables with 50–90% missing values that were judged uninformative, and grouped moderately correlated items, guided by SHAP-value clustering from a LightGBM model [18, 19]. The LightGBM step used the outcomes and a split of the full cohort that does not correspond to the evaluation splits. Because all models received the same 105 predictors, this step could raise the performance of all models

but does not favour any one of them. [VERIFY: whether the SHAP step removed or retained any variable — E-04] Responses of "don't know" or "don't want to answer" were set to missing. [VERIFY: grouped bullying/abuse items use −1 for missing (feature_selection.py:116-130) — missing_code_check.csv, A-03]

2.4 Data partitioning

The data were split at the twin-pair level, so that co-twins were always in the same partition, into 75% training and 25% test sets. This was repeated with five random seeds (42–46). The same five splits were used for every model: the split identifiers were exported once and read by the R scripts. [VERIFY: split_summary.csv reports matches_dcmfnet_log = True for all seeds] Each test set contained 998–1,028 individuals. For DCMFNet, a further pair-level 80/20 split of each training set formed a validation set for early stopping and learning-rate scheduling.

2.5 Deep cross-modal fusion network

We adapted DCMFNet [9], originally proposed for fusing image and text representations through iterative gated fusion (IGF), to grouped tabular predictors (Fig. 2). Substance use at age 15 is the anchor group. Each of groups 2–10 is fused with the anchor in its own IGF block, and a parallel independent stream receives all raw inputs, including the batch × PC terms. [TODO: Fig. 2 architecture diagram regenerated from code (C-04)]

Fusion block. Let $\mathbf{x}_a$ be the anchor features ($d_a = 6$) and $\mathbf{x}_m$ the d_m features of group m. Block m has depth L_m and starts from $\mathbf{g}_m^{(0)} = \mathbf{x}_m$. For $l = 1, \ldots, L_m$:

$$\mathbf{f}_m^{(l)} = \tanh\big(\tanh(\mathbf{W}_1^{(l)}\mathbf{x}_a + \mathbf{b}_1^{(l)}) \odot \tanh(\mathbf{W}_2^{(l)}\mathbf{g}_m^{(l-1)} + \mathbf{b}_2^{(l)})\big), \tag{1}$$

$$\mathbf{z}_m^{(l)} = \sigma\big(\mathbf{W}_z^{(l)}(\mathbf{f}_m^{(l)} + \mathbf{g}_m^{(l-1)}) + \mathbf{b}_z^{(l)}\big), \tag{2}$$

$$\mathbf{g}_m^{(l)} = \tanh\big(\mathbf{W}_g^{(l)}[\mathbf{f}_m^{(l)};\, \mathbf{f}_m^{(l)} \odot \mathbf{z}_m^{(l)}] + \mathbf{b}_g^{(l)}\big), \tag{3}$$

where σ is the logistic function, $\odot$ element-wise multiplication and $[\cdot;\cdot]$ concatenation. Equation 1 is a multiplicative interaction between the anchor and the current representation of group m; Equations 2–3 form the gate of the original architecture. The block output concatenates the fused representations from all layers, $\mathbf{u}_m = [\mathbf{f}_m^{(1)}; \ldots; \mathbf{f}_m^{(L_m)}]$.

Feature gating. Each block output, the concatenated fused stream, the independent stream and the final representation pass through a feature gate

$$\phi(\mathbf{u}) = \mathbf{u} \odot \sigma\big(\mathbf{V}_2\,\mathrm{Dropout}(\mathrm{ReLU}(\mathbf{V}_1\mathbf{u} + \mathbf{c}_1)) + \mathbf{c}_2\big), \tag{4}$$

with hidden width $\max(\lfloor D/r \rfloor, h_{\min})$ for input dimension D. This gate was motivated by squeeze-and-excitation attention [20], but it is computed for each individual from that individual's own features, without pooling. It is therefore an input-dependent weighting, not a population-level importance measure.

Streams and output. The fused stream is $\mathbf{h}_F = \phi_F([\phi_1(\mathbf{u}_1); \ldots; \phi_9(\mathbf{u}_9)])$ and the independent stream is $\mathbf{h}_I = \phi_I(\mathbf{x})$, where $\mathbf{x}$ contains all 105 inputs. The prediction is $\hat{y} = \mathbf{w}^\top \phi_{\text{out}}([\mathbf{h}_F; \mathbf{h}_I]) + b$, with a linear output. Because the independent stream feeds gated raw inputs to a linear output layer, the network contains an approximately additive pathway alongside the interaction pathway. Inputs were not standardised.

Loss. To give more weight to the sparse upper tail of the outcomes, we used a weighted Huber loss. Training outcomes were assigned to K equal-width bins on [0, 1]; each bin received a weight proportional to $1/(n_k + 1)$, normalised to mean 1 and truncated at 20. Within each mini-batch, the weights were multiplied by a focal factor $(|y_i - \hat{y}_i| + 10^{-6})^\gamma$, computed without gradient, and renormalised to mean 1.

Training and configuration selection. Networks were trained with the Adam optimiser [CITATION NEEDED: Kingma & Ba 2015], weight decay, learning-rate reduction on plateau of the validation RMSE and early stopping with restoration of the best weights. Hyperparameters were selected with Optuna [21] (50 trials per outcome), minimising validation RMSE within the seed-42 training set (Additional file 1: Table S1). The search tuned one depth for all fusion blocks (best: 3 for PLE-HM, 2 for DEP). A subsequent sweep varied the depth of each block. For PLE-HM, the per-block depths from this sweep were adopted after per-block and uniform configurations had been compared on the test sets; for DEP, uniform depth 2 was kept. Several tuning runs were carried out, and the final choice among them was also informed by test-set results. In addition, the seed-42 training data used for tuning overlap the test sets of seeds 43–46. All of these steps can only make DCMFNet's test performance optimistic relative to the comparators, which were not tuned. The selected hyperparameters are given in Table 2. [AUTHOR INPUT: confirm that the per-block PLE-HM configuration [5,2,1,5,4,1,1,1,5] is the primary one — D-05, M-02]

Table 1: Selected DCMFNet hyperparameters.

Hyperparameter	PLE-HM model	DEP model
Learning rate	1.00×10^{-4}	3.22×10^{-4}
Batch size	4	6
Maximum epochs	23	20
Weight decay	2.95×10^{-5}	1.79×10^{-4}
Fusion depth L_m (PRS, SCZ15, ADHD9, ASD9, ACE15, ACE18, SUD18, SES, Sex)	5, 2, 1, 5, 4, 1, 1, 1, 5	2 (all blocks)
Dropout	0.234	0.326
Gate reduction ratio r / minimum width $h_{\min}$	6 / 13	5 / 5
Huber δ / focal γ / bins K	0.086 / 1.32 / 13	0.116 / 2.62 / 10
Scheduler patience / factor	4 / 0.432	2 / 0.549
Early-stopping patience	5	7
Trainable parameters	105,501	71,623

2.6 Mixed-effects comparators

Linear mixed model (LMM). The LMM included all 105 predictors as linear main effects, a random intercept for twin pair and 24 interaction terms that mirror the network's fusion blocks: the SUD15 score multiplied by the score of each fused questionnaire group (SCZ15, ADHD9, ASD9, ACE15, ACE18, SUD18 and SES; 7 terms), by sex (1 term) and by each standardised PRS (16 terms). Group scores were the mean of the standardised items, standardised again; means and standard deviations were estimated in the training set and applied to the test set. The model was fitted with lme4::lmer (Gaussian, identity link). In the code and output files this model is labelled "GLMM". [AUTHOR INPUT: use "LMM" in the paper? The thesis used "GLMM" for a Gaussian model.]

Generalised additive mixed model (GAMM). The GAMM had the same specification, except that each PRS entered as a penalised smooth term and the twin-pair intercept as a random-effect smooth. It was fitted with mgcv::gam (Gaussian family, restricted maximum likelihood). [VERIFY: basis dimension (mgcv default k = 10, thin-plate) and mgcv version]

For both models, test-set predictions were made at the population level, without the twin-pair random effect, because test pairs were not part of the training data.

Machine-learning baselines. Forty regression algorithms, including penalised linear, kernel, nearest-neighbour, tree-ensemble, boosting and neural network models, were fitted with default settings using LazyPredict [22] on the same predictors and splits. [VERIFY: LazyPredict version and internal preprocessing — A-05] These models did not account for twin pairs, and estimators with internal cross-validation (for example LassoCV) did not group it by twin pair. [VERIFY]

2.7 Evaluation and statistical comparison

All models were evaluated on the same five test sets with the same metric code: the coefficient of determination, $R^2 = 1 - \sum(y_i - \hat{y}_i)^2 / \sum(y_i - \bar{y})^2$, root-mean-squared error (RMSE), mean absolute error (MAE) and Spearman's rank correlation. Results are given as mean and standard deviation over the five splits.

Models were compared split by split with the corrected resampled t-test, which accounts for the overlap between training sets across splits [23] (test-to-training ratio 1/3, 4 degrees of freedom). P-values were adjusted with the Holm method within each metric. With five splits these tests have low power, so a non-significant difference is not evidence of equivalence.

For the mixed models, interaction terms were assessed with their Wald p-values, Holm-adjusted across the 24 terms, and PRS smooth terms by their effective degrees of freedom.

2.8 Calibration, subgroup and sensitivity analyses

The comparison was repeated with participant-level test predictions from DCMFNet retrained with the same seeds, splits and hyperparameters, so that all models were scored with identical code.

- **Calibration** [24]: calibration-in-the-large (mean observed minus mean predicted), calibration slope and intercept from regressing observed on predicted values, and calibration curves over deciles of the predictions.
- **Upper tail:** within the 10% of test participants with the highest observed scores, we report the precision of the model's top 10% of predictions, the area under the ROC curve for identifying this group, and RMSE and mean bias.
- **Sex:** all metrics computed separately for males and females.
- **Predictors measured by age 15:** all models were refitted without ACE18 and SUD18. The mixed models dropped these variables and their two interaction terms (97 main effects, 22 interactions). For DCMFNet, whose architecture expects every group, both inputs were set to their training-set mean in training and test data, which removes their information while keeping the tuned architecture.

2.9 Attribution analyses

We asked whether DCMFNet's gate values reflect what the network relies on for its predictions, and whether DCMFNet and the LMM rely on the same predictor groups. For each split, the following quantities were computed for the 11 predictor groups on the test set, from the same retrained models:

1. **Gate values:** the mean gate value per group in DCMFNet's fused and independent streams, as reported in the thesis.
2. **SHAP values for DCMFNet** [19]: expected-gradients SHAP values (100 training rows as background, 200 samples per explained row), summed within each group; group importance was the mean absolute group SHAP value.
3. **Exact SHAP values for the LMM:** for the population-level prediction, with the training data as reference. Main effects contribute $\beta_j(x_j - \bar{x}_j)$. Each product term γSP was split between SUD15 and its partner according to its two-player Shapley values, $\frac{\gamma}{2}(S - \bar{S})(P + \bar{P})$ and $\frac{\gamma}{2}(P - \bar{P})(S + \bar{S})$.
4. **Grouped permutation importance** for both models: the drop in test R^2 when all variables of a group are permuted jointly, averaged over 10 permutations.

Agreement between sources was summarised by Spearman's rank correlation across groups and the overlap of the three highest-ranked groups, and stability by the correlation between splits. All attributions describe associations used by the fitted models, not causal effects.

2.10 Software and reporting

Analyses used Python [VERIFY: version] with PyTorch, scikit-learn, Optuna, SHAP and LazyPredict [VERIFY: versions], and R [VERIFY: version] with lme4 and mgcv [VERIFY: versions]. Code is available at [TODO: tagged release with DOI — C-05]. Reporting follows the TRIPOD+AI statement [25] (Additional file 2). [TODO: complete the checklist — C-06]

[AUTHOR INPUT: Springer Nature asks that use of large language models be documented in the Methods (copy-editing excepted) — VERIFY: JOURNAL REQUIREMENT. Draft: "A large language model (Claude, Anthropic) was used to

audit the analysis code against the reported methods, to write analysis scripts that the authors reviewed and ran, and to draft and edit text. The authors checked all outputs and take full responsibility for the content."]

2.11 Ethics

[AUTHOR INPUT: approving committee(s), reference number(s) and date(s) for CATSS and the genotype linkage; consent procedure (parents for minors, participants at age 18); statement that the study was carried out in accordance with relevant guidelines and regulations / the Declaration of Helsinki — TODO C-02] The same information is given under *Declarations*.

3 Results

3.1 Analysis sample

The analysis sample comprised 4,033 individuals from 2,690 twin pairs with complete data on all 105 predictors and both outcomes. Each of the five test sets held 25% of the twin pairs, between 998 and 1,028 individuals. [TODO: Table 1, analysis sample vs excluded (A-02), and Fig. 1, participant flow (W-02) — from analysis_sample_descriptives.R]

3.2 Predictive performance

All models were evaluated on the same five twin-pair-grouped train/test splits. [VERIFY: split_summary.csv reports matches_dcmfnet_log = True for all seeds] R^2 is 1 − SSE/SST throughout, and SDs are across the five splits (population SD).

Table 2: Test-set performance, mean ± SD over five splits.

Model	Outcome	R^2	RMSE	MAE	Spearman ρ
DCMFNet	PLE-HM	0.2058 ± 0.0189	0.0903 ± 0.0013	0.0712 ± 0.0004	0.4497 ± 0.0193
GLMM	PLE-HM	0.2279 ± 0.0156	0.0890 ± 0.0014	0.0700 ± 0.0012	0.4641 ± 0.0183
GAMM	PLE-HM	0.2274 ± 0.0153	0.0891 ± 0.0014	0.0699 ± 0.0011	0.4633 ± 0.0180
DCMFNet	DEP	0.1984 ± 0.0208	0.1562 ± 0.0017	0.1242 ± 0.0019	0.4509 ± 0.0135
GLMM	DEP	0.2025 ± 0.0220	0.1559 ± 0.0024	0.1226 ± 0.0016	0.4460 ± 0.0154
GAMM	DEP	0.1998 ± 0.0230	0.1561 ± 0.0026	0.1228 ± 0.0017	0.4421 ± 0.0163

Mean R^2 was higher for the GLMM and GAMM than for DCMFNet on both outcomes. The difference was 0.022 for PLE-HM and 0.004 for DEP (GLMM minus

DCMFNet). GLMM and GAMM performed almost identically. For DEP, DCMFNet had the highest Spearman correlation (0.451 vs 0.446 and 0.442).

Penalised linear models fitted with default settings on the same splits did better than all three models. The cross-validated lasso had mean R^2 0.261 (SD 0.019) for PLE-HM and 0.234 (0.017) for DEP (Additional file 1: Table S2).

[TODO: add the default-setting ML baselines (same splits as DCMFNet) as a supplementary table; the best linear baselines had mean R^2 0.261 (LassoCV, PLE-HM) and 0.235 (LarsCV, DEP) — Models/ML/Results/{Pos,Neg}_benchmark_summary.csv. Decide with the author whether they appear in Table 3.]

3.3 Paired comparisons across splits

The models were compared split by split with the corrected resampled t-test (Nadeau and Bengio), using the per-split metrics, with Holm correction across the six comparisons per metric (three model pairs × two outcomes) [23].

Table 3: Paired differences in test R^2 (first model minus second), five splits.

Outcome	Comparison	Mean ΔR^2 (SD)	Splits where first model was better	p	p (Holm)
PLE-HM	DCMFNet − GLMM	−0.0221 (0.0133)	1 of 5	0.086	0.50
PLE-HM	DCMFNet − GAMM	−0.0216 (0.0129)	1 of 5	0.083	0.50
PLE-HM	GLMM − GAMM	0.0005 (0.0007)	4 of 5	0.40	1.00
DEP	DCMFNet − GLMM	−0.0041 (0.0253)	2 of 5	0.84	1.00
DEP	DCMFNet − GAMM	−0.0013 (0.0270)	2 of 5	0.95	1.00
DEP	GLMM − GAMM	0.0027 (0.0025)	4 of 5	0.22	0.87

None of the differences was statistically significant. RMSE gave the same pattern: PLE-HM DCMFNet − GLMM +0.0013, p = 0.085; DEP +0.0004, p = 0.84. With five splits these tests have little power, so the results show no detectable advantage for DCMFNet. They do not show that the models are equivalent. [TODO: replace with prediction_evaluation.py paired_comparisons.csv once run (retrained DCMFNet, identical metric code; adds MAE, Spearman, calibration slope)]

3.4 Interaction and smooth terms in the mixed models

Coefficients come from the fits on the first split (seed 42). The GLMM and GAMM each included 24 SUD15 interaction terms: - SUD15 score × each fused modality score (7); - SUD15 score × sex; - SUD15 score × each of the 16 cell-type PRS.

- **PLE-HM.** Three of the 24 interactions had unadjusted $p < 0.05$ in both the GLMM and the GAMM: SUD15 × the PRS for medium spiny neurons, MGE interneurons and upper-layer intratelencephalic neurons ($|\beta|$ 0.005–0.006, p 0.018–0.038).
- **DEP.** One interaction had unadjusted $p < 0.05$: SUD15 × SCZ15 ($\beta = -0.006$, $p = 0.017$ in the GLMM).
- **Multiple testing.** About 1.2 terms in 24 would reach $p < 0.05$ by chance alone. None of these terms remains significant after correction for 24 tests (smallest Holm-adjusted $p \geq 0.40$).
- We therefore found no robust evidence of SUD15 interactions in the statistical models.

In the GAMM, 11 of the 16 PRS smooths were close to linear for each outcome (effective degrees of freedom < 1.1). The maximum was 2.54 for PLE-HM and 1.92 for DEP. Smooths with unadjusted $p < 0.05$: - **PLE-HM:** the all-cell PRS and the PRS for hippocampal CA1–3 and miscellaneous cells; - **DEP:** the all-cell PRS only.

[TODO: full coefficient table as Additional file (all terms, not only $p < 0.05$)]

3.5 Attribution: gates, SHAP and permutation importance

[TODO: E-06 — from shap_analysis.py, GLMM_shap_updated.R and attribution_agreement.py]

3.6 Calibration, subgroup and sensitivity analyses

[TODO: E-09 calibration, A-04 sex-stratified, E-10 top-decile, E-05 age-15-only predictors — from prediction_evaluation.py]

4 Discussion

In a population-based twin cohort, a deep cross-modal fusion network built around adolescent substance use did not predict psychotic-like and hypomanic experiences or depressive symptoms at age 18 better than linear and additive mixed models that were given the same predictors and interaction terms mirroring the network's fusion design. Mean test R^2 was about 0.02 lower for the network than for the mixed models for PLE-HM and similar for DEP, and no paired difference was statistically significant. Penalised linear models fitted with default settings did best. The mixed models also showed little evidence of the substance-use interactions that motivate the fusion design: the number of nominally significant interaction terms was close to the number expected by chance, and most PRS effects were close to linear. [TODO: one or two sentences on calibration and the age-15-only analysis, from prediction_evaluation.py] [TODO: one sentence on whether gates agreed with SHAP and permutation importance, from attribution_agreement.py]

These findings are consistent with comparisons in clinical prediction, where flexible machine-learning models rarely outperform regression when the comparison is at low risk of bias [10], and with benchmarks showing that deep networks are seldom the best choice for medium-sized tabular data [11]. They also agree with earlier machine-learning work in this cohort [12] [VERIFY: content]. Two features of our design make the result conservative for the network. First, only the network was tuned, and part of its configuration was chosen with test-set information; both can only favour it. Second, the mixed models were given the same interaction structure as the network, so the comparison isolates how the interactions are modelled rather than whether they are included. [TODO: L-02 — compare with direct competitors (ML/DL with PRS and environmental predictors for psychotic or depressive outcomes) once identified]

Several explanations are possible. The training sets contained about 3,000 individuals, while the networks had 72,000 to 106,000 trainable parameters (Table 2). At this sample size, flexible models are likely to fit noise rather than signal [26] [VERIFY: content]. The predictors and outcomes are questionnaire sum scores and PRS, which explain a limited share of variance, and this limits the attainable R^2 for any model. Finally, if interactions between substance use and other domains are weak at the population level, a model designed to capture them has little to gain. Our results cannot distinguish between these explanations. They also do not show that such interactions are absent: prediction accuracy is a different question from testing moderation, and the interaction tests here had limited power after correction for multiple testing.

[TODO — attribution paragraph (E-06). Report: (a) agreement between DCMFNet gate values and SHAP / permutation importance on the same models (Spearman, top-3 overlap); (b) agreement between DCMFNet and the LMM; (c) seed stability. If gates disagree with SHAP/permutation, conclude that gate values should not be read as predictor importance [@R-JAIN], as was done in the thesis. If they agree, say that in this setting gate rankings were consistent with model-agnostic attributions, but that this does not make them causal.]

[TODO — calibration, sex and age-15 paragraph (E-09, A-04, E-05, E-10). Report calibration slopes and calibration-in-the-large for each model; whether conclusions differ by sex; how much performance drops without ACE18/SUD18 and whether the model ranking changes; upper-tail performance.]

4.1 Strengths and limitations

Strengths of this study include a population-based twin cohort with genetic and multi-informant questionnaire data; evaluation of all models on identical twin-pair-grouped splits with the same metric code; comparators that matched the network's inputs and interaction structure; paired inference that accounts for overlapping training sets; and analyses of calibration, sex, predictors measured before the outcome and attribution.

The study also has limitations.

- **Configuration selection.** The network's depth configuration, and the choice among tuning runs, used test-set results, and the tuning data overlapped later test sets. The network's own performance estimates are therefore likely optimistic, although this bias works against our main conclusion.

- **Feature screening.** The variable-screening step used the outcomes on data that overlap the test sets [13]. This affects all models equally but may inflate absolute performance.
- **Complete-case analysis.** Only individuals with complete data were analysed, about 44% of the eligible sample [VERIFY: against analysis_sample_flow.csv]. [TODO: summarise Table 1 differences between included and excluded participants]. Results may not generalise to the full cohort.
- **Outcome measures.** The outcomes are continuous questionnaire scores, not diagnoses, and the PLE-HM score combines psychotic-like and hypomanic items. [AUTHOR INPUT: M-01 instrument details]
- **Concurrent predictors.** In the main analysis, adverse experiences and substance use at age 18 were measured at the same time as the outcomes. [TODO: state how results changed when they were removed]
- **Twin dependence.** The mixed models included a twin-pair random intercept, but the network and the machine-learning baselines did not model twin dependence during training; splits were grouped by twin pair for all models.
- **Generalisability.** A single cohort with no external validation; PRS derived from European-ancestry genome-wide association studies [VERIFY: GWAS ancestry — W-03].
- **Power.** Five splits give limited power to detect small differences between models.

5 Conclusions

In this Swedish twin cohort, a deep fusion network designed around interactions between adolescent substance use and other risk domains did not predict symptoms at age 18 better than mixed-effects models with matched inputs and interactions, and penalised linear models did best. The data gave little evidence of the interactions that motivated the network. [TODO: one sentence on attribution agreement once available] Deep models for psychiatric risk prediction in cohort data should be compared with well-specified regression models on identical, family-aware splits, with model selection kept away from the test data, before their added complexity or their attention-based explanations are relied on.

Abbreviations. ACE: adverse childhood experiences; ADHD: attention-deficit/hyperactivity disorder; ASD: autism spectrum disorder; CATSS: Child and Adolescent Twin Study in Sweden; DCMFNet: Deep Cross-Modal Fusion Network; DEP: depressive symptoms; GAMM: generalised additive mixed model; IGF: iterative gated fusion; LMM: linear mixed model; MAE: mean absolute error; PC: principal component; PLE-HM: psychotic-like and hypomanic experiences; PRS: polygenic risk score; RMSE: root-mean-squared error; SES: socioeconomic status; SHAP: Shapley additive explanations; SUD: substance use; TRIPOD+AI: Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis, updated for artificial intelligence.

Supplementary information.

- **Additional file 1** (PDF): Table S1, Optuna search space; Table S2, all 40 machine-learning baselines; Table S3, full LMM and GAMM coefficients; Table S4, variable list by predictor group; Table S5, depth-configuration results (5 seeds); tuning history (D-21). [TODO]
- **Additional file 2** (PDF): TRIPOD+AI checklist. [TODO: C-06]

Declarations

Ethics approval and consent to participate. [AUTHOR INPUT: "CATSS and the genotype linkage were approved by [committee, e.g., the Regional Ethical Review Board in Stockholm / Swedish Ethical Review Authority], reference number(s) [xxx], date(s) [xxx]. [Consent procedure: parents for participants under 15/18; participants themselves at age 18 — VERIFY]. The study was conducted in accordance with the Declaration of Helsinki."] — TODO C-02

Consent for publication. Not applicable.

Availability of data and materials. Draft: "The CATSS data contain sensitive personal information and cannot be shared publicly under Swedish data-protection law. Researchers can apply for access through the Swedish Twin Registry, Karolinska Institutet [AUTHOR INPUT: exact procedure/URL]. The analysis code is available at [TODO: repository release + DOI — C-05]."

Competing interests. [AUTHOR INPUT] or: "The authors declare that they have no competing interests."

Funding. [AUTHOR INPUT: funders and grant numbers for CATSS, genotyping and this work, and the role of the funders] or: "This research received no specific funding."

Authors' contributions. [AUTHOR INPUT] Example structure only: "XX designed the study, wrote the analysis code, analysed the data and drafted the manuscript. YY supervised the study and contributed to the design and interpretation. All authors read and approved the final manuscript."

Acknowledgements. [AUTHOR INPUT: CATSS participants and staff; colleagues (names only with their consent)]

Clinical trial number. Not applicable. [VERIFY: JOURNAL REQUIREMENT]

Figure legends

Fig. 1 Participant flow from the CATSS cohort to the complete-case analysis sample. [TODO: from analysis_sample_flow.csv — W-02]

Fig. 2 Architecture of the deep cross-modal fusion network. Substance use at age 15 (anchor) is fused with each of nine predictor groups in iterative gated fusion blocks (fused stream); an independent stream receives all inputs. [TODO: diagram regenerated from code — C-04]

Fig. 3 Calibration curves for each model and outcome, by decile of predicted score, pooled over the five test sets. [TODO: calibration_curves_full.pdf from prediction_evaluation.py]

Fig. 4 Predictor-group importance by source: DCMFNet gate values, DCMFNet SHAP, DCMFNet permutation importance, LMM SHAP and LMM permutation importance. [TODO: from attribution_agreement.py]

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